GRAVEATLAS

Phase 14 — Global Intelligence, Digital Twin & Predictive Preservation

GRAVEATLAS — PHASE 14
GLOBAL INTELLIGENCE, DIGITAL TWIN & PREDICTIVE PRESERVATION

PHASE 1 THROUGH PHASE 13 MUST ALREADY BE COMPLETE.

Continue from the existing GraveAtlas project.

CORE RULES
- Do NOT rebuild Phases 1–13.
- Do NOT start Phase 15.
- Reuse existing architecture, security, provenance, moderation, audit, preservation, APIs, datasets, automation and release systems.
- No paid services or external integrations without explicit approval.
- Prefer existing/free/local processing where practical.
- Never expose secrets, credentials, tokens or private data.
- Never fabricate historical facts, records, institutions, users, analytics, coverage, performance or predictive results.
- Never silently alter authoritative historical records.
- Predictions and AI suggestions are NOT historical facts.
- Do not perform destructive or mass operations without authorization.
- Complete Phase 14 only, then stop.

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PART 1 — PHASE 13 BASELINE
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Read the actual Phase 13 implementation and final report.

Identify:
- completed capabilities
- remaining blockers
- automation maturity
- monitoring maturity
- AI limitations
- data-quality limitations
- preservation risks
- security/privacy constraints

Use evidence only.

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PART 2 — DIGITAL TWIN ARCHITECTURE
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Evaluate or implement a safe digital representation of the GraveAtlas ecosystem.

Represent where appropriate:
- datasets
- cemeteries
- records
- sources
- people
- places
- organizations
- maps
- APIs
- indexes
- preservation systems
- automation
- dependencies

The digital twin must represent actual system state.
Never fabricate entities to fill gaps.

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PART 3 — SYSTEM STATE MODEL
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Where practical track:
- availability
- data health
- provenance health
- preservation health
- indexing health
- automation health
- security state
- dependency state

State must be derived from actual observations.
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PART 4 — PREDICTIVE PRESERVATION
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Identify potential future preservation risks such as:
- obsolete formats
- broken source dependencies
- disappearing source links
- incomplete metadata
- missing provenance
- storage growth
- migration requirements
- single points of failure

Predictions must clearly state:
- evidence
- assumptions
- uncertainty
- recommended action

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PART 5 — PRESERVATION RISK PRIORITY
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Where useful classify risks:
CRITICAL
HIGH
MEDIUM
LOW
MONITOR

Do not create arbitrary scores without a documented methodology.

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PART 6 — HISTORICAL KNOWLEDGE GRAPH
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Evaluate or improve relationships between:
- people
- records
- cemeteries
- places
- events
- sources
- organizations

Every relationship must preserve:
- evidence
- provenance
- confidence/uncertainty
- source references

Do not convert inference into fact.

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PART 7 — ADVANCED GEOGRAPHIC INTELLIGENCE
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Where supported by actual data, analyze:
- historical cemetery locations
- cemetery boundaries
- geographic changes
- former locations
- historical map relationships
- regional patterns

Every geographic claim must remain traceable to its source.

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PART 8 — TEMPORAL INTELLIGENCE
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Provide safe time-based exploration for:
- cemetery history
- record chronology
- source chronology
- historical events
- geographic changes

Clearly distinguish:
CONFIRMED
INFERRED
UNKNOWN
DISPUTED

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PART 9 — KNOWLEDGE DISCOVERY

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Use actual GraveAtlas data to identify:
- potentially related records
- unconnected sources
- duplicate candidates
- evidence gaps
- research opportunities

AI may recommend connections but must not silently create authoritative relationships.

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PART 10 — PREDICTIVE DATA QUALITY
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Where justified, predict potential:
- duplicate clusters
- suspicious records
- high-risk datasets
- recurring quality problems
- future moderation workload

Predictions must remain reviewable.

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PART 11 — PRESERVATION DEPENDENCY ANALYSIS
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Map dependencies between:
- records
- sources
- external references
- storage
- indexes
- APIs
- datasets
- preservation copies

Identify single points of failure.

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PART 12 — INSTITUTIONAL INTELLIGENCE
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Where authorized, provide institutional views of:
- collection health
- verification status
- preservation status
- data quality
- provenance
- research activity

Do not expose private institutional information.

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PART 13 — GLOBAL RESEARCH GRAPH
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Support navigation such as:

PERSON
→ RECORD
→ CEMETERY
→ LOCATION
→ SOURCE
→ INSTITUTION
→ HISTORICAL EVENT

Every edge must remain evidence-aware.

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PART 14 — EXPLAINABLE PREDICTIONS
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Every predictive recommendation must provide:
- prediction
- evidence
- reasoning/explanation
- confidence or uncertainty
- limitations
- recommended action

Do not present predictions as certainty.

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PART 15 — HUMAN GOVERNANCE

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Predictive systems must recommend rather than silently decide historical truth.

Authorized humans must be able to:
- review predictions
- reject recommendations
- approve valid changes
- quarantine suspicious data
- override automation
- inspect evidence

All important decisions must be auditable.

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PART 16 — LONG-TERM PLATFORM SIMULATION
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Safely model potential:
- dataset growth
- storage growth
- API growth
- search growth
- map growth
- preservation requirements
- infrastructure requirements

Use documented assumptions.

Do not represent projections as actual measurements.

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PART 17 — GLOBAL INTELLIGENCE DASHBOARD
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Where justified, provide authorized dashboards for:
- data health
- preservation risk
- geographic coverage
- provenance health
- institutional status
- automation health
- security status
- research opportunities

Protect internal operational information.

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PART 18 — AI GOVERNANCE
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Track AI-assisted predictive work:
- model/provider
- version where available
- workflow/prompt version
- source inputs
- generated output
- confidence
- human review
- final decision

Never send private information to external AI without approval.

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PART 19 — PREDICTION SAFETY
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Do not use predictive systems to:
- invent historical records
- declare uncertain relationships as facts
- fabricate sources
- automatically rewrite authoritative history
- infer sensitive personal information unnecessarily

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PART 20 — SECURITY
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Review:
- knowledge graph access
- prediction services
- dashboards
- AI pipelines
- institutional data
- APIs
- automation credentials

- stored predictions

Apply least privilege and secure defaults.

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PART 21 — PRIVACY
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Review privacy implications of:
- relationship graphs
- research activity
- institutional analytics
- predictive systems
- user behavior
- AI processing

Collect only necessary information.

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PART 22 — PERFORMANCE
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Safely evaluate:
- graph size
- search performance
- map performance
- prediction workloads
- dashboard performance
- storage requirements

Use non-production testing where possible.

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PART 23 — FAILURE & RECOVERY
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Every predictive or intelligence workflow must define:
- failure detection
- timeout
- retry limits
- fallback
- escalation
- audit logging

Do not create uncontrolled autonomous loops.

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PART 24 — DOCUMENTATION
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Create/update:

docs/DIGITAL-TWIN.md
docs/SYSTEM-STATE.md
docs/PREDICTIVE-PRESERVATION.md
docs/PRESERVATION-RISK.md
docs/HISTORICAL-KNOWLEDGE-GRAPH.md
docs/GEOGRAPHIC-INTELLIGENCE.md
docs/TEMPORAL-INTELLIGENCE.md
docs/KNOWLEDGE-DISCOVERY.md
docs/PREDICTIVE-DATA-QUALITY.md
docs/DEPENDENCY-ANALYSIS.md
docs/INSTITUTIONAL-INTELLIGENCE.md
docs/GLOBAL-RESEARCH-GRAPH.md
docs/EXPLAINABLE-PREDICTIONS.md
docs/PREDICTION-GOVERNANCE.md
docs/PLATFORM-SIMULATION.md
docs/GLOBAL-INTELLIGENCE-DASHBOARD.md
docs/PHASE-14-ROADMAP.md

Document actual implementation only.

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PHASE 14 ACCEPTANCE GATE
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[ ] Phase 13 baseline reviewed
[ ] Digital twin architecture
[ ] System state model
[ ] Predictive preservation
[ ] Preservation risk analysis
[ ] Historical knowledge graph
[ ] Geographic intelligence
[ ] Temporal intelligence
[ ] Knowledge discovery
[ ] Predictive data quality

[ ] Dependency analysis
[ ] Institutional intelligence
[ ] Global research graph
[ ] Explainable predictions
[ ] Human governance
[ ] Platform simulation
[ ] Global intelligence dashboard
[ ] AI governance
[ ] Prediction safety
[ ] Security review
[ ] Privacy review
[ ] Performance evaluation
[ ] Failure/recovery controls
[ ] Documentation

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FINAL REPORT
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PHASE 14 STATUS: READY / NOT READY

For every implemented capability report:
- what was built
- evidence
- tests performed
- limitations
- security considerations
- privacy considerations
- provenance considerations
- operational impact

Use actual metrics only.

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PREDICTION SCORECARD
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Digital twin:
[actual assessment]

Preservation prediction:
[actual assessment]

Knowledge graph:
[actual assessment]

Geographic intelligence:
[actual assessment]

Temporal intelligence:
[actual assessment]

Data-quality prediction:
[actual assessment]

Explainability:
[actual assessment]

Human oversight:
[actual assessment]

Security:
[actual assessment]

Do not invent numeric scores without a defined methodology.

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BLOCKERS
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List every issue preventing Phase 14 from being READY.

For each:
- severity
- component
- issue
- evidence
- impact
- remediation
- status

Never include secrets.

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ROADMAP
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NOW:
[items]

NEXT:
[items]

LATER:
[items]

DO NOT BUILD YET:
[items]

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STOP CONDITION
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Do NOT start Phase 15.

Do NOT add paid services without explicit approval.

Do NOT expose secrets.

Do NOT fabricate predictions, data, sources or historical relationships.

Do NOT silently modify authoritative historical records.

Do NOT publish private or restricted information.

Do NOT perform destructive bulk operations.

Do NOT perform uncontrolled production load testing.

Do NOT treat AI predictions as historical truth.

STOP after the Phase 14 final report.

WAIT FOR EXPLICIT INSTRUCTIONS FOR PHASE 15.